Assignment 4

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Problem 1

a-)

```
CDI.data <- read.csv("/cloud/project/Data/CDI Data.csv")
Y <- CDI.data$Number.of.active.physicians
X1 <- CDI.data$Total.population
X2 <- CDI.data$Total.personal.income
X3 <- I(CDI.data$Geographic.region == 1)*1 # NE
X4 <- I(CDI.data$Geographic.region == 2)*1 # NC
X5 <- I(CDI.data$Geographic.region == 3)*1 # S
CDI.data2 <- data.frame(cbind(Y,X1,X2,X3,X4,X5))
CDI.reg <- lm(Y ~ ., data=CDI.data2)
summary(CDI.reg)
```

```
##
## Call:
## lm(formula = Y ~ ., data = CDI.data2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1866.8  -207.7   -81.5    72.4  3721.7
##
## Coefficients:
##                 Estimate   Std. Error t value           Pr(>|t|)
## (Intercept) -207.4957492   70.2816094  -2.952            0.00332 **
## X1             0.0005515    0.0002835   1.945            0.05243 .
## X2             0.1070115    0.0132550   8.073 0.0000000000000068 ***
## X3           149.0195716   86.8331592   1.716            0.08685 .
## X4           145.5264460   85.1529259   1.709            0.08817 .
## X5           191.2163044   80.0330139   2.389            0.01731 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 566.1 on 434 degrees of freedom
## Multiple R-squared:  0.9011, Adjusted R-squared:  0.8999
## F-statistic: 790.7 on 5 and 434 DF,  p-value: < 0.00000000000000022
```

```
n <- nrow(CDI.data2) # 440
p <- 6
b3 <- CDI.reg$coefficients[4] # 149.02
b4 <- CDI.reg$coefficients[5] # 145.53
T.crit <- qt(0.975,n-p) # 1.965
b3.se <- summary(CDI.reg)$coefficients[4,2] # 86.833
```

```r
b4.se <- summary(CDI.reg)$coefficients[5,2] # 85.153
b3.lwr <- b3 - T.crit*b3.se # 5.89
b3.upr <- b3 + T.crit*b3.se # 292.15
b4.lwr <- b4 - T.crit*b4.se # 5.16
b4.upr <- b4 + T.crit*b4.se # 285.89

# Confirm the results with 'confint' function
confint(CDI.reg)[4:5,]
```

```
##        2.5 %   97.5 %
## X3 -21.64623 319.6854
## X4 -21.83695 312.8898
```

Confidence Intervals (calculated manually):
-21.646 $\leq \beta_3 \leq$ 319.685
-21.837 $\leq \beta_4 \leq$ 312.89

The confidence intervals for β_3 and $\beta 4$ are very similar and mostly overlap with each other, so the effect for NE region on number of active physicians does not differ from the effect for NC region.

$*$ Below, I attempt to create the confidence interval for $\beta_4 - \beta_3$

```r
b4b3 <- b4 - b3 # -3.493
b4.b3.se <- vcov(CDI.reg)[5,4] # 4289.97
b4.var <- vcov(CDI.reg)[5,5] # 7251.02
b3.var <- vcov(CDI.reg)[4,4] # 7540
b4b3.var <- b4.var + b3.var - 2*b4.b3.se # 6211.09
b4b3.lwr <- b4b3 - T.crit*sqrt(b4b3.var) # -158.39
b4b3.upr <- b4b3 + T.crit*sqrt(b4b3.var) # 151.40
```

$(b_4 - b_3) \pm t(1 - \frac{\alpha}{2}, df)s\{b_4 - b_3\}$
$s^2\{b_4 - b_3\} = s^2\{b_4\} + s^2\{b_3\} - 2s\{b_4, b_3\}$

Confidence Interval: -158.391 $\leq \beta_4 - \beta_3 \leq$ 151.404

The confidence interval above includes zero, so we cannot conclude that the effect for the NE region on number of physicians differs from that for NC region. It is consistent with the result from comparing two confidence intervals above.

b-)

```r
# Reduced model
CDI.reg.reduced <- lm(Y ~ X1+X2, data=CDI.data2)
SSE_F <- anova(CDI.reg)$`Sum Sq`[6]
SSE_R <- anova(CDI.reg.reduced)$`Sum Sq`[3]
dfF <- 434 # n - 6
dfR <- 437 # n - 3
F.stat <- ((SSE_R-SSE_F)/(dfR-dfF)) / (SSE_F/dfF) # 1.949
F.crit <- qf(0.9,dfR-dfF,dfF) # 2.096
pv <- 1-pf(F.stat,3,434) # 0.121
```

Full Model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5$
Reduced Model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$
H_a: not all β_3 and β_4 and β_5 equal zero

$F^* = \frac{SSE(R)-SSE(F)}{df_R - df_F} \div \frac{SSE(F)}{df_F} = 1.949$
$F(1-\alpha; df_R - df_F, df_f) = \mathrm{F}(0.95;3,434) = 2.096$

$F^* \leq F(0.95; 3, 434)$, so we conclude H_0 that no geographic effects are present. p-value = 0.121.

Below is the simpler method to get the F^* and p-value.

```
anova(CDI.reg, CDI.reg.reduced)
```

```
## Analysis of Variance Table
##
## Model 1: Y ~ X1 + X2 + X3 + X4 + X5
## Model 2: Y ~ X1 + X2
##   Res.Df       RSS Df Sum of Sq      F Pr(>F)
## 1    434 139093455
## 2    437 140967081 -3  -1873626 1.9487  0.121
```

$F^* = 1.949$ and p-value = 0.121 are the same as the one we calculated manually.

Problem 2

a-)

```
GR.data <- read.csv("/cloud/project/Data/Grocery Retailer Data Set.csv")
GR.reg <- lm(Y ~ X1+X3+X2, data=GR.data)
summary(GR.reg)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X3 + X2, data = GR.data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -264.05 -110.73  -22.52   79.29  295.75
##
## Coefficients:
##                 Estimate   Std. Error t value             Pr(>|t|)
## (Intercept) 4149.8872120  195.5654055  21.220 < 0.0000000000000002 ***
## X1             0.0007871    0.0003646   2.159               0.0359 *
## X3           623.5544807   62.6409466   9.954    0.000000000000294 ***
## X2           -13.1660192   23.0917315  -0.570               0.5712
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 143.3 on 48 degrees of freedom
## Multiple R-squared:  0.6883, Adjusted R-squared:  0.6689
## F-statistic: 35.34 on 3 and 48 DF,  p-value: 0.000000000003316
```

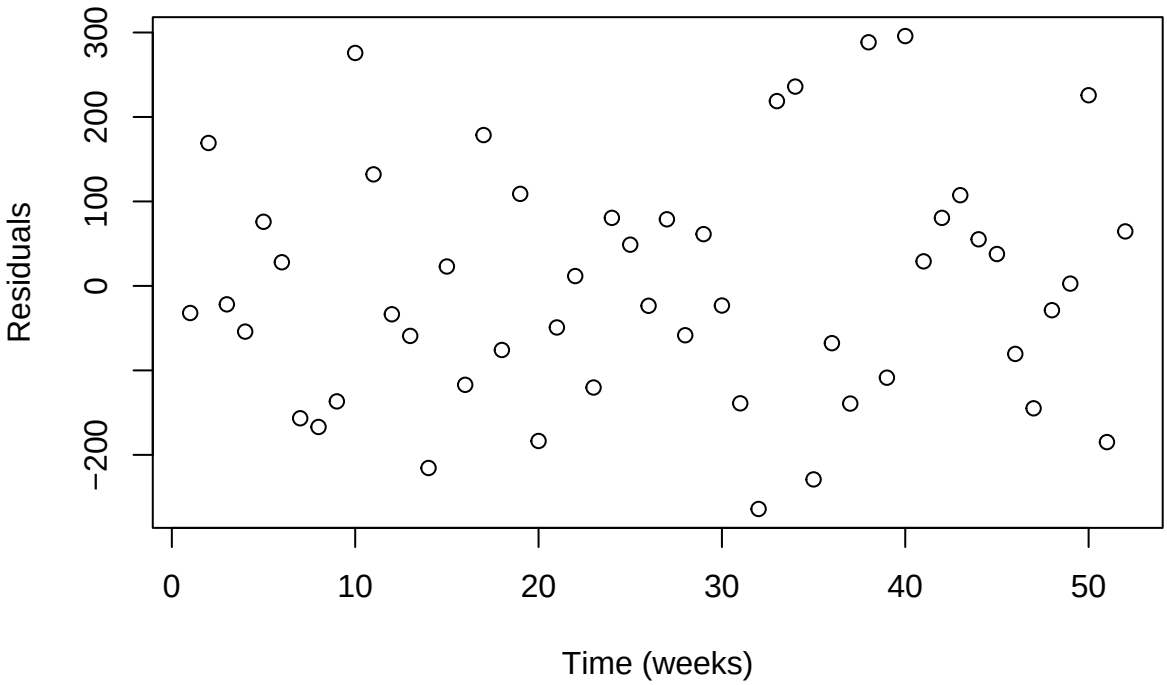
```
b0 <- GR.reg$coefficients[1] # 4,149.89
b1 <- GR.reg$coefficients[2] # 0.000787
b2 <- GR.reg$coefficients[4] # -13.166
b3 <- GR.reg$coefficients[3] # 623.55
```

The estimated regression function: Y = 4149.887 + 0.001X_1 + -13.166X_2 + 623.554X_3

The interpretation of the coefficients:
b_1: the change in E[Y] with a unit change in X_1 while X_2 and X_3 are held constant
b_2: the change in E[Y] with a unit change in X_2 while X_1 and X_3 are held constant
b_3: the change in E[Y] with a unit change in X_3 while X_1 and X_2 are held constant

b-)

```
plot(GR.reg$residuals, xlab="Time (weeks)", ylab="Residuals")
```



The time plot of residuals shown above seems to have no pattern and shows no indication that the error terms are correlated.

c-)

```
anova(GR.reg)
```

```
## Analysis of Variance Table
##
## Response: Y
##           Df  Sum Sq Mean Sq F value             Pr(>F)
## X1         1  136366  136366  6.6417            0.01309 *
## X3         1 2033565 2033565 99.0443 0.0000000000002963 ***
## X2         1    6675    6675  0.3251            0.57123
## Residuals 48  985530   20532
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
SSR.X1 <- anova(GR.reg)$`Sum Sq`[1] # 136,366
SSR.X3_1 <- anova(GR.reg)$`Sum Sq`[2] # 2,033,565
SSR.X2_13 <- anova(GR.reg)$`Sum Sq`[3] # 6,675
```

$SSR(X_1) = 136366$
$SSR(X_3|X_1) = 2033565$
$SSR(X_2|X_1, X_3) = 6675$

d-)

```
GR.reg.reduced <- lm(Y ~ X1+X3, data=GR.data)
anova(GR.reg, GR.reg.reduced)
```

```
## Analysis of Variance Table
```

```
##
## Model 1: Y ~ X1 + X3 + X2
## Model 2: Y ~ X1 + X3
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     48 985530
## 2     49 992204 -1   -6674.6 0.3251 0.5712
```

```
F1 <- anova(GR.reg, GR.reg.reduced)[2,5] # F = 0.3251
F.crit <- qf(0.95,1,48) # 4.04
pvalue <- 1 - pf(F1,1,48) # 0.5712
```

$H_0 : \beta_2 = 0$ $H_a : \beta_2 \neq 0$

ANOVA table shows that $F^* = 0.3251 \leq$ F(0.95;1,48) $= 4.04$, so we conclude H_0 that X_2 can be dropped from the regression model that already contains X_1 and X_3. The p-value is 0.571 and is greater than the significant level (α=0.05).

$*$Alternative way to get F^*

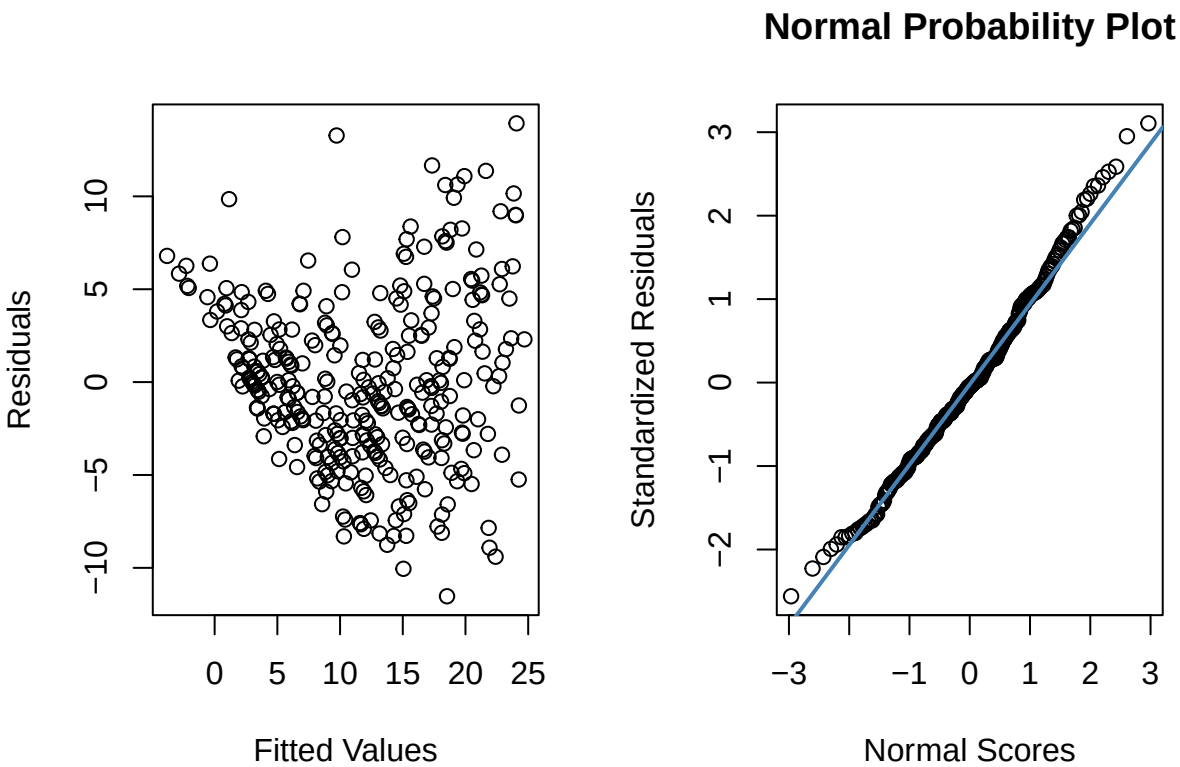
```
F2 <- anova(GR.reg)$`Mean Sq`[3] / anova(GR.reg)$`Mean Sq`[4]
```

$F^* = \frac{MSR(X_2|X_1,X_3)}{MSE(X_1,X_2,X_3)}$ (p.264 of textbook)
$F^* = 0.3251$

Problem 3

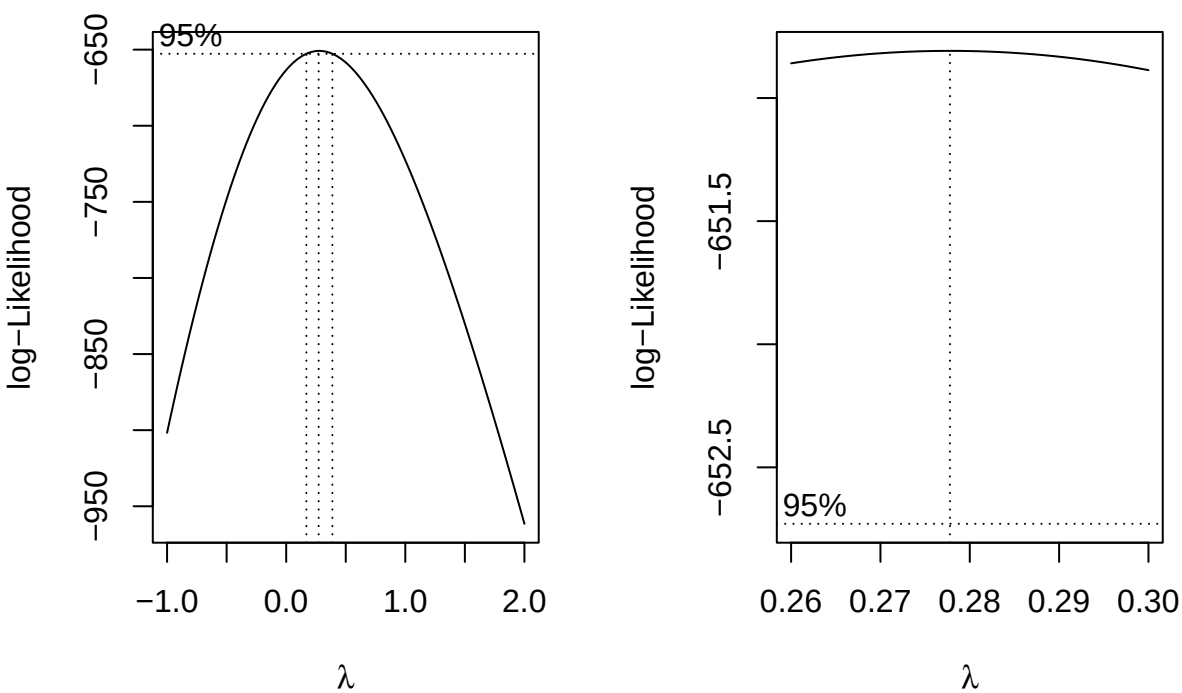
```
data(ozone)
ozone.reg <- lm(O3 ~ temp+humidity+ibh, data=ozone)

par(mfrow=c(1,2))
plot(ozone.reg$fitted.values, ozone.reg$residuals, xlab="Fitted Values", ylab="Residuals")
error.std3 <- rstandard(ozone.reg)
qqnorm(error.std3, xlab="Normal Scores", ylab="Standardized Residuals",
       main="Normal Probability Plot")
qqline(error.std3, col="steelblue", lwd=2)
```



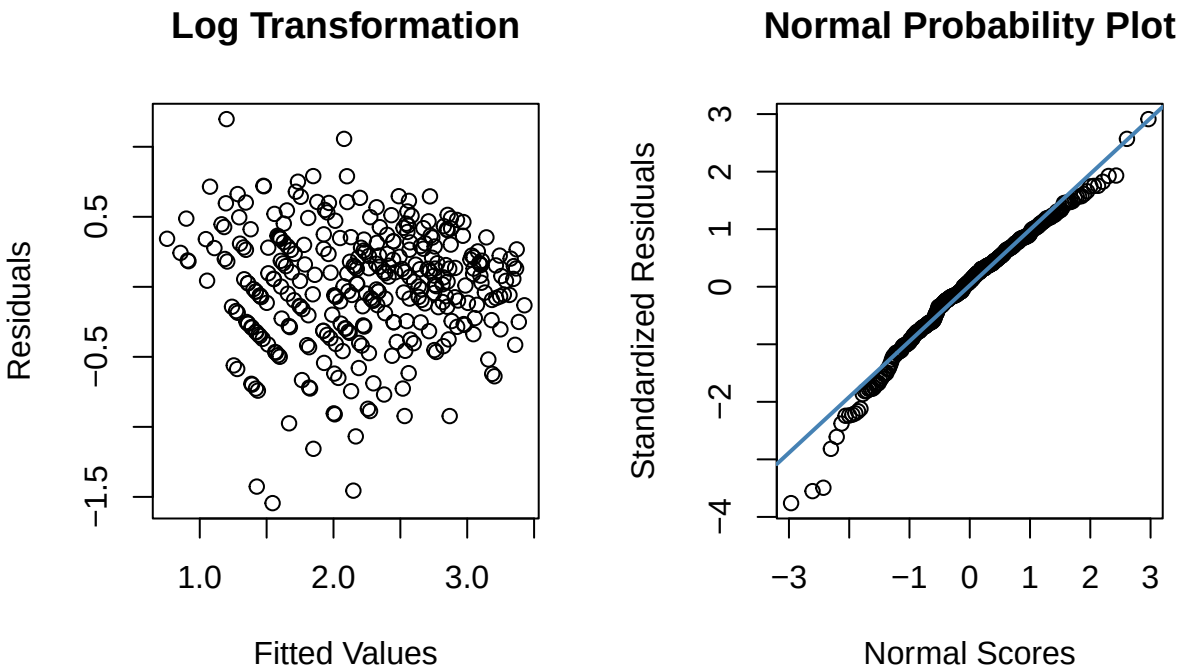
QQ plot shows that the error terms are approximately normally distributed, but the residuals vs. fitted values plot indicates the nonconstancy of the error terms. As the fitted values increase, the variability of residuals also increases a lot. So, we will try to find a proper transformation using the Box-Cox procedure below.

```
par(mfrow=c(1,2))
boxcox(ozone.reg,lambda=seq(-1,2,by=.1))
boxcox(ozone.reg,lambda=seq(0.26,0.3,by=.01))
```

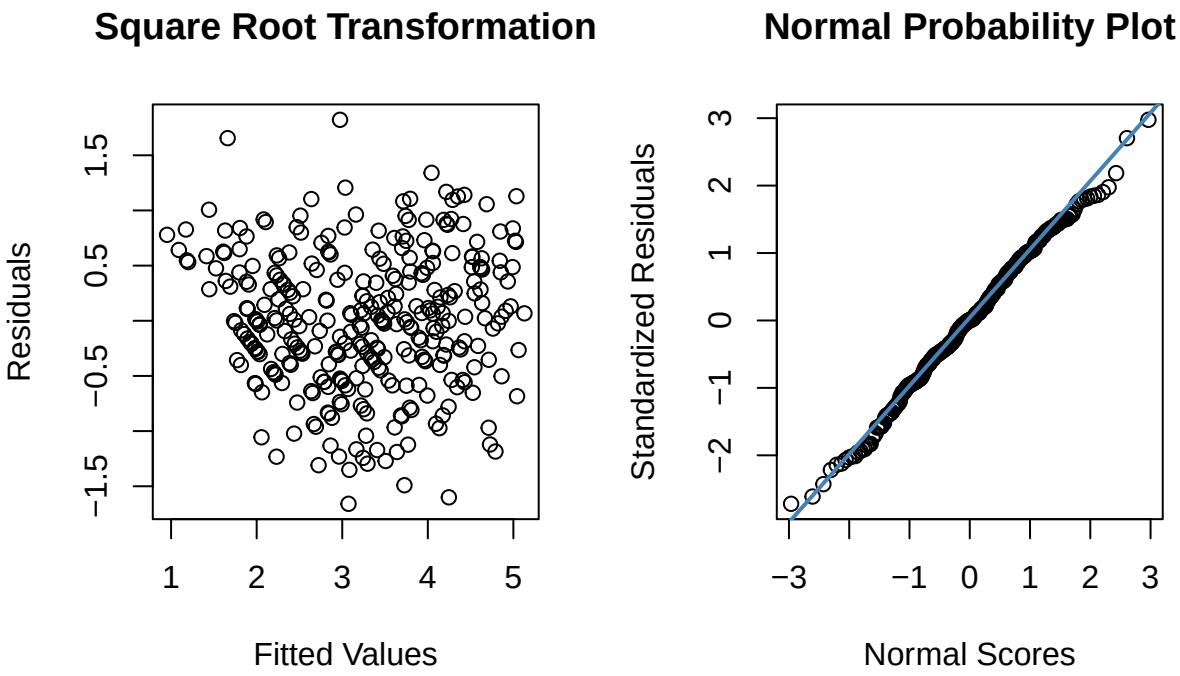


The zoom-in version of the Box-Cox plot on the right shows that λ is around 0.28. Since λ=0 suggests the logarithmic transformation and λ=0.5 suggests the square root transformation of Y, it is not very clear which of the two is more appropriate in our case. Maybe either one is justifiable. Further analysis seems necessary.

```
par(mfrow=c(1,2))
ozone.reg.log <- lm(log(O3) ~ temp+humidity+ibh, data=ozone)
plot(ozone.reg.log$fitted.values, ozone.reg.log$residuals, xlab="Fitted Values", ylab="Residuals",
     main="Log Transformation")
error.std4 <- rstandard(ozone.reg.log)
qqnorm(error.std4, xlab="Normal Scores", ylab="Standardized Residuals",
       main="Normal Probability Plot")
qqline(error.std4, col="steelblue", lwd=2)
```




```
par(mfrow=c(1,2))
ozone.reg.sqrt <- lm(sqrt(O3) ~ temp+humidity+ibh, data=ozone)
plot(ozone.reg.sqrt$fitted.values, ozone.reg.sqrt$residuals, xlab="Fitted Values", ylab="Residuals",
     main="Square Root Transformation")
error.std5 <- rstandard(ozone.reg.sqrt)
qqnorm(error.std5, xlab="Normal Scores", ylab="Standardized Residuals",
       main="Normal Probability Plot")
qqline(error.std5, col="steelblue", lwd=2)
```




```
sh1 <- shapiro.test(ozone.reg$residuals)$p.value
sh2<- shapiro.test(ozone.reg.log$residuals)$p.value
sh3 <- shapiro.test(ozone.reg.sqrt$residuals)$p.value
tbl2 <- data.frame("Original"=sh1, "Log Transformation"=sh2, "Sqrt Transformation"=sh3)
kable(tbl2, format="latex", digits=6, caption="Shapiro-Wilk test p-values") %>%
  kable_styling(latex_options="HOLD_position")
```

Table 1: Shapiro-Wilk test p-values

Original	Log.Transformation	Sqrt.Transformation
0.067141	0.00009	0.551333

```
r2 <- summary(ozone.reg)$r.squared
r2.log <- summary(ozone.reg.log)$r.squared
r2.sqrt <- summary(ozone.reg.sqrt)$r.squared
tbl <- data.frame("R-squared"=r2, "R-squared (log)"=r2.log, "R-squared (Sqrt)"=r2.sqrt)
kable(tbl, format="latex", digits=4, caption="Comparison of R-squared Values") %>%
  kable_styling(latex_options="HOLD_position")
```

Table 2: Comparison of R-squared Values

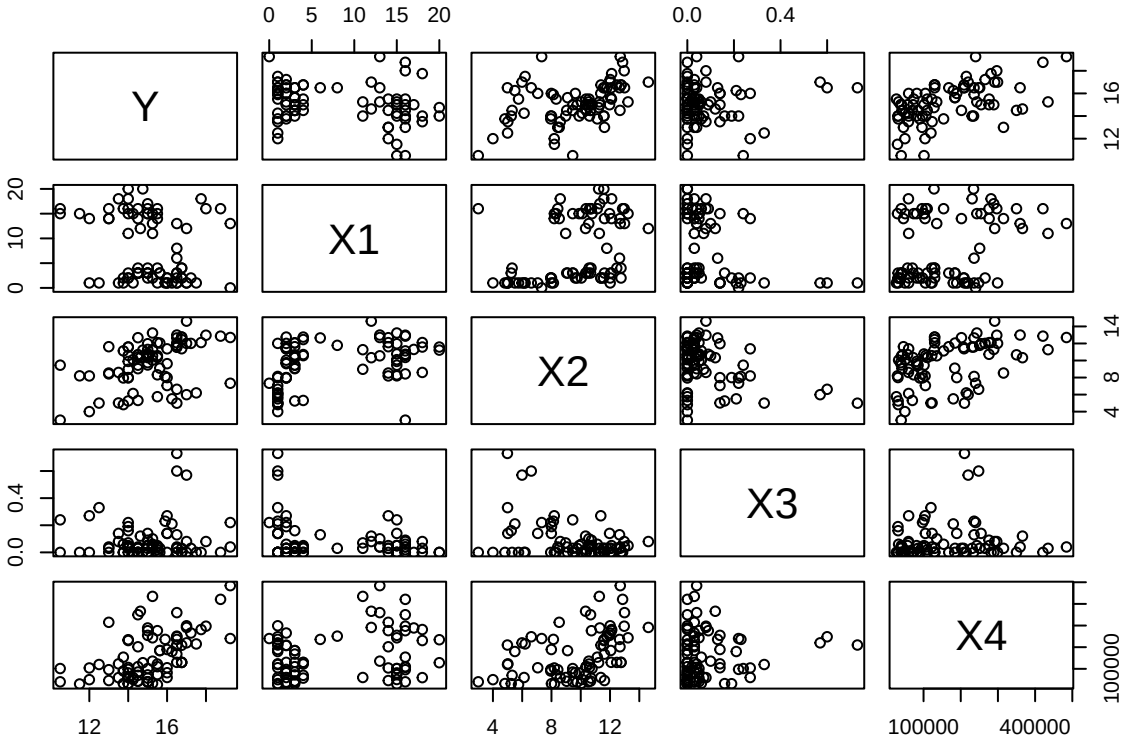
R.squared	R.squared..log.	R.squared..Sqrt.
0.684	0.6975	0.717

The QQ plot of the log transformed model has a curvy shape, and the Shapiro-Wilk test result shows that it is not normally distributed whereas the normality holds for the square root transformation. The residuals plots show that the square root transformation is better than the log transformation, showing more constancy of variability. Also, R^2 is the highest for the square root transformation despite the small difference. Given above results, we would choose the square root transformation, but some additional adjustment might be necessary because there seems to be significantly lower variances of residuals for lower fitted values.

Problem 4

a-)

```
CP.data <- read.csv("/cloud/project/Data/Commercial Properties Data Set.csv")
plot(CP.data)
```




```
round(cor(CP.data),2)
```

```
##        Y    X1    X2    X3   X4
## Y   1.00 -0.25  0.41  0.07 0.54
## X1 -0.25  1.00  0.39 -0.25 0.29
## X2  0.41  0.39  1.00 -0.38 0.44
## X3  0.07 -0.25 -0.38  1.00 0.08
## X4  0.54  0.29  0.44  0.08 1.00
```

The scatter plot matrix and correlation matrix above indicate that each of the predictor variables is linearly associated with the response variable (Y) with X_4 showing the highest degrees of association and X_3 the lowest. The matrices also show the intercorrelations among the predictor variables. In particular, X_2 has some degree of pairwise correlations with other predictor variables while X_3 and X_4 are almost uncorrelated.

b-)

```
CP.reg <- lm(Y ~ X1+X2+X3+X4, data=CP.data)
summary(CP.reg)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X2 + X3 + X4, data = CP.data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.1872 -0.5911 -0.0910  0.5579  2.9441
##
```

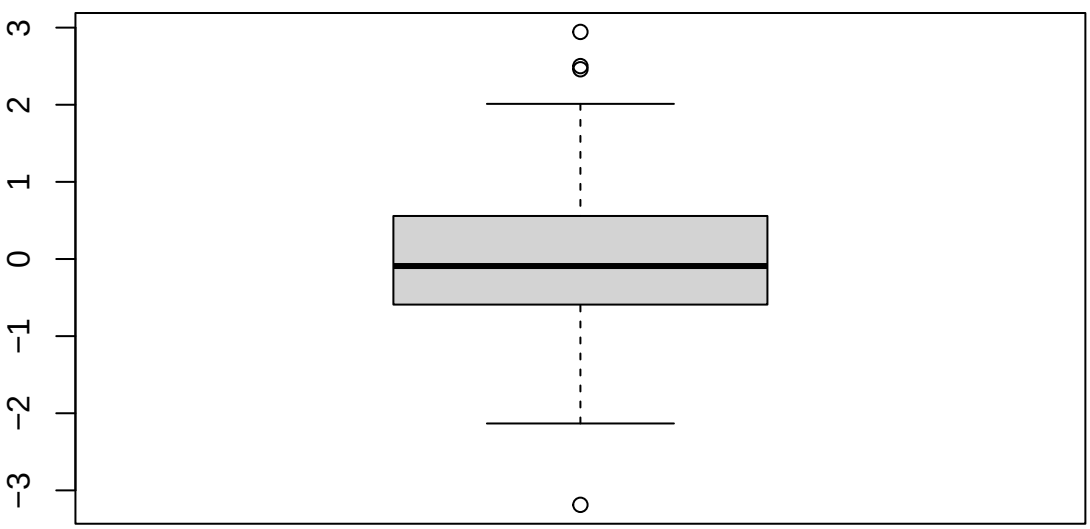
```
## Coefficients:
##                 Estimate   Std. Error t value             Pr(>|t|)
## (Intercept) 12.200585882  0.577956174  21.110 < 0.0000000000000002 ***
## X1          -0.142033644  0.021342610  -6.655        0.00000000389 ***
## X2           0.282016530  0.063172350   4.464        0.00002747396 ***
## X3           0.619343503  1.086812829   0.570                 0.57
## X4           0.000007924  0.000001385   5.722        0.00000019760 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.137 on 76 degrees of freedom
## Multiple R-squared:  0.5847, Adjusted R-squared:  0.5629
## F-statistic: 26.76 on 4 and 76 DF,  p-value: 0.00000000000007272
```

```r
b0 <- CP.reg$coefficients[1] # 12.2
b1 <- CP.reg$coefficients[2] # -0.142
b2 <- CP.reg$coefficients[3] # 0.282
b3 <- CP.reg$coefficients[4] # 0.619
b4 <- CP.reg$coefficients[5] # 0.0000079
```

The estimated regression function: Y = 12.201 - 0.142X_1 + 0.282X_2 + 0.619X_3 + 0.0000079X_4

c-)

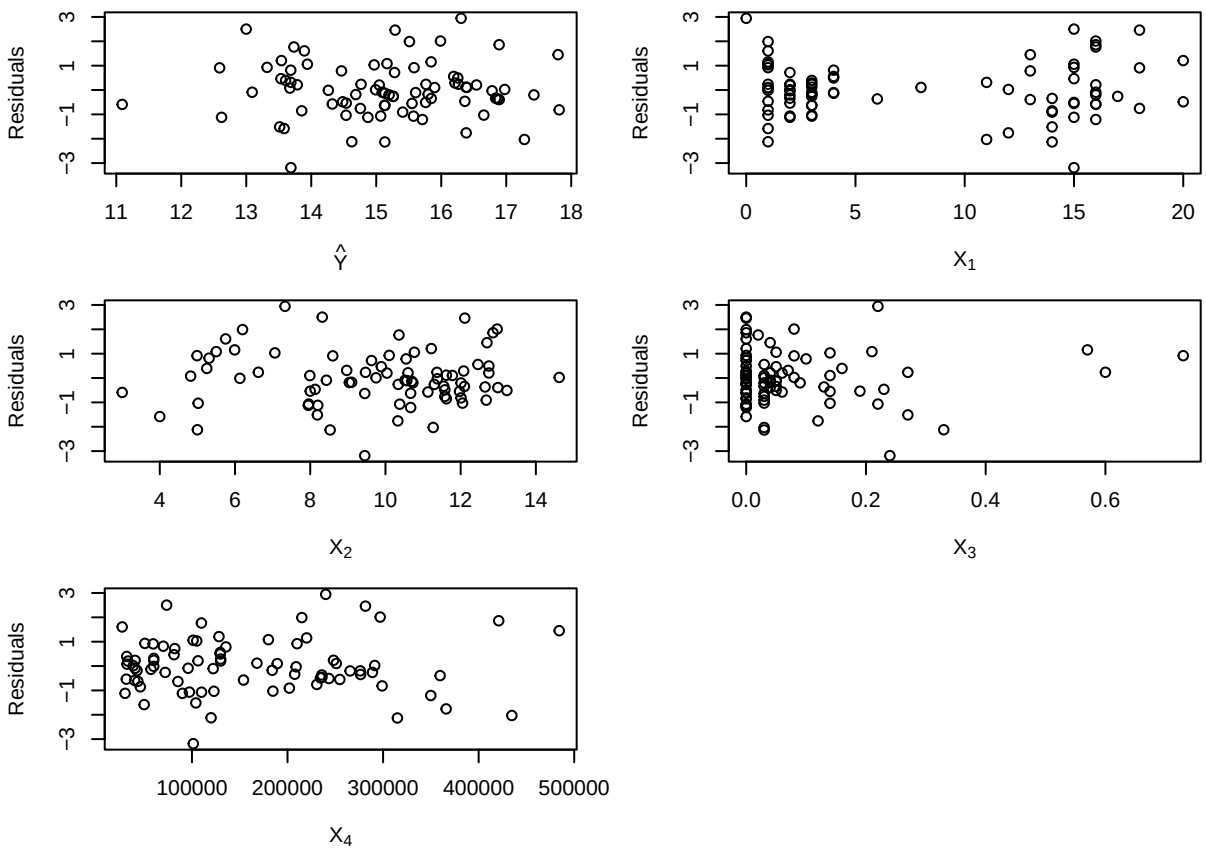
```r
ei <- CP.reg$residuals
boxplot(ei)
```



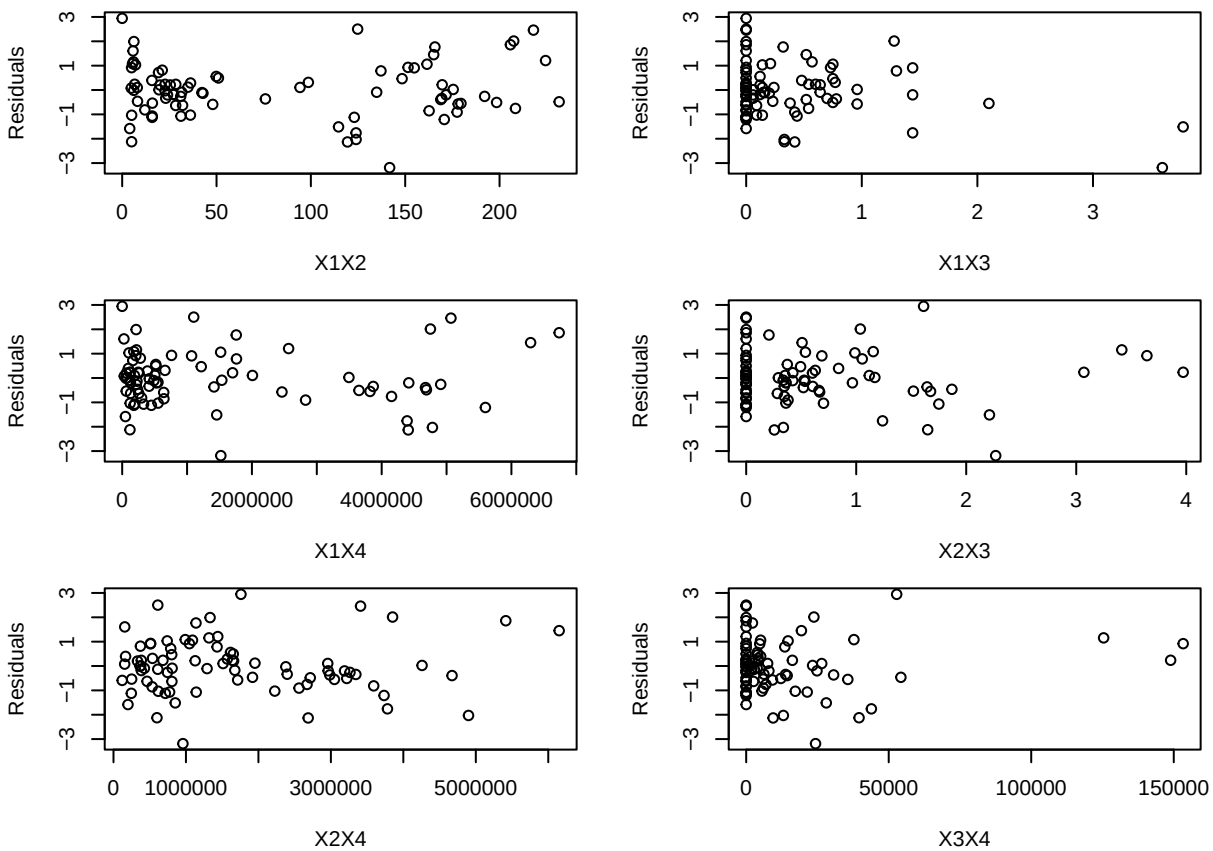
The box plot of the residuals shows a fairly symmetrical distribution.

d-)

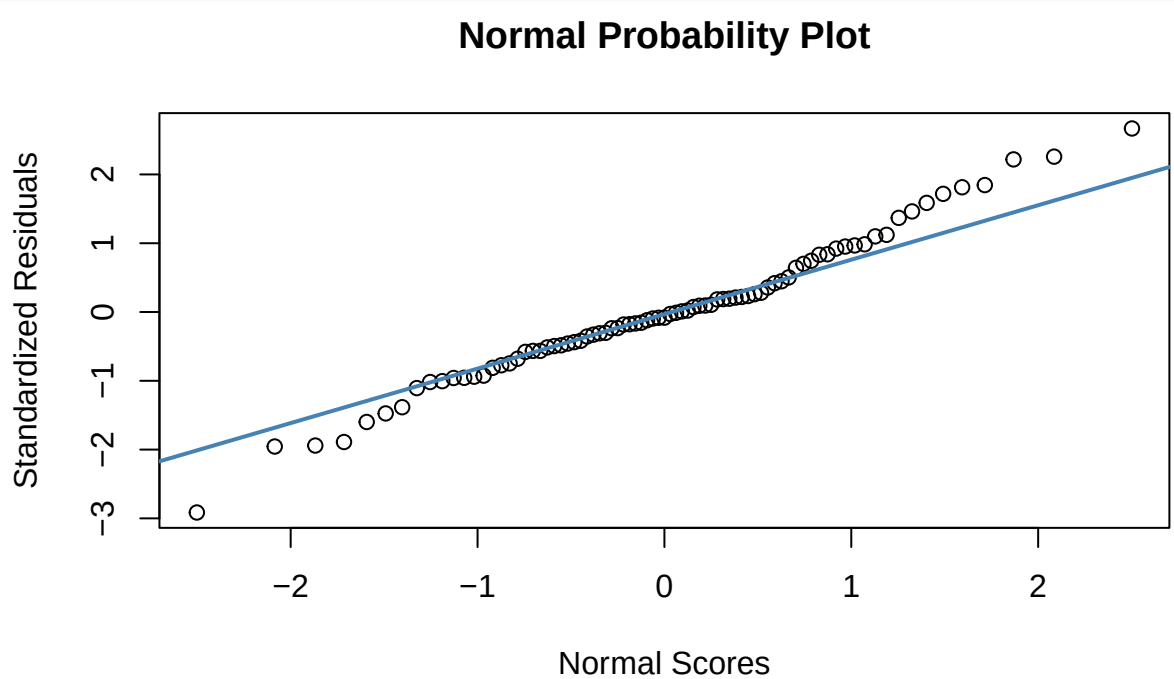
```r
par(mfrow=c(3,2), mar=c(4,4,1,2))
plot(CP.reg$fitted.values, ei, xlab=expression(hat(Y)), ylab="Residuals")
plot(CP.data$X1, ei, xlab=expression(X[1]), ylab="Residuals")
plot(CP.data$X2, ei, xlab=expression(X[2]), ylab="Residuals")
plot(CP.data$X3, ei, xlab=expression(X[3]), ylab="Residuals")
plot(CP.data$X4, ei, xlab=expression(X[4]), ylab="Residuals")
par(mfrow=c(3,2), mar=c(4,4,1,2))
```




```
plot(CP.data$X1*CP.data$X2, ei, xlab="X1X2", ylab="Residuals")
plot(CP.data$X1*CP.data$X3, ei, xlab="X1X3", ylab="Residuals")
plot(CP.data$X1*CP.data$X4, ei, xlab="X1X4", ylab="Residuals")
plot(CP.data$X2*CP.data$X3, ei, xlab="X2X3", ylab="Residuals")
plot(CP.data$X2*CP.data$X4, ei, xlab="X2X4", ylab="Residuals")
plot(CP.data$X3*CP.data$X4, ei, xlab="X3X4", ylab="Residuals")
```




```
par(mfrow=c(1,1))
error.std <- rstandard(CP.reg)
qqnorm(error.std, xlab="Normal Scores", ylab="Standardized Residuals",
       main="Normal Probability Plot")
qqline(error.std, col="steelblue", lwd=2)
```



All the residuals plots do not seem to show any significant pattern. Some of them have points concentrated on the left side near zero of the x-axis, but it is caused by the differences of scale in the predictor variables. For example, 30 data points of X_3 are zero, so the two-factor interaction terms including X_3 will also have 30 zeros. The normal probability plot indicates that the error terms are approximately normally distributed, slightly heavy-tailed.

e-)

```
bf.data <- data.frame(cbind(CP.data, ei, CP.reg$fitted.values))
dimnames(bf.data)[[2]][6:7] <- c("residuals","fitted.values")
bf.data1<-data.frame(cbind(
  bf.data,ind=as.factor(I(bf.data$fitted.values<median(bf.data$fitted.values))*1)))
g1 <- sum(bf.data1$ind=="1") # group 1: 40 cases
g2 <- sum(bf.data1$ind=="0") # group 2: 41 cases
bf.test(residuals~ind, data=bf.data1)
```

```
##
##   Brown-Forsythe Test (alpha = 0.05)
## -------------------------------------------------------------
##   data : residuals and ind
##
##   statistic  : 1.421309
##   num df     : 1
##   denom df   : 78.46183
##   p.value    : 0.2367813
##
##   Result     : Difference is not statistically significant.
## -------------------------------------------------------------
```

Following the Brown-Forsythe test result shown above, we conclude H_0 that the error variances are constant.